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MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAM MARCH 2024

Course: B. Tech Semester: IV

Subject Name: Electronic Devices and Circuits Subject Code: EE 223

Time Duration: 1Hour 30 Minutes Maximum Marks: 20

S No	Question	Marks	CO
1.(a)	Identify and Justify the application of Op-Amp, if inverting amplifier is modified by (i) R_f is replaced by **C** and R_i is replaced by **C**.	2.5	CO2
1.(b)	Realize an Op-Amp circuit to solve a differential equation of $$\frac{d^2v}{dt^2} = -40\frac{dv}{dt} - 100v + X$$ (X = Last two digits of Sch. Number)	2.5	CO3
2.(a)	How feedback system affects the performance of Op-Amp? Design circuit for generation of 650 Hz sinusoidal waveform using Wein-bridge oscillator. Assume **C= 0.05 μF and R1 = 10 kΩ**	2.5	CO3
2.(b)	Estimate the ON time, OFF time and free running frequency of output wave form, if 555 timer is working in astable mode. Given **V_{cc} = 5 V, R_A=2.2 k Ω, R_B= 3.9 k Ω and C = 0.1 μF.**	2.5	CO4
3.	How half adder and half subtractor circuit can be implemented using logic gates with solution of K- Maps ?	05	CO3
4.	Design logic gate circuit with K- Maps solution for operation of BCD to 7 segment decoder to display "7". (Assume common anode display) *(i) Using IC* *(ii) Using basic logic gates*	05	CO4